

A mobility-based framework for measuring socio-spatial segregation in 30 U.S. metropolitan areas

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ABSTRACT

Although “segregation” is widely used and often treated as a clear condition, what it actually means for a place to be “highly segregated” remains ambiguous. Studies invoke the term across diverse social and spatial contexts, from income to race, or from residence to facility access, without a shared definition of what segregation entails. This ambiguity obscures comparisons across cities and dimensions, risks misdiagnosing the sources of inequality, and undermines targeted interventions. To address this conceptual and empirical gap, we propose a socio-spatial dimensional framework that defines segregation as the intersection between social dimensions (who differs) and spatial axes (how that difference is structured and experienced). The social dimensions correspond to the Social Vulnerability Index (SVI), including Socioeconomic Status (SES), Household Composition & Disability (HCD), Minority Status & Language (MSL), and Housing Type (HT). The spatial axes capture metropolitan unevenness and local clustering. Using mobility data from 45 million mobile devices, we construct this mobility-based framework, Social Segregation Index (SSI), for the 30 largest U.S. metropolitan areas. Results show that “high segregation” has no single meaning: rankings of metropolitan areas change dramatically by social dimensions, with severity and clustering strongest along SES and MSL but weaker for HCD and HT. Extreme multi-axis segregation is rare yet concentrated in a few urban cores. By clarifying who differs and how in space, our framework transforms the vague notion of segregation into a measurable, comparable, and policy-relevant socio-spatial condition.

1. Introduction

“Segregation” is a deceptively simple label applied to complex socio-spatial realities from the very beginning (e.g., Duncan & Duncan, 1955; Park, 1925; White, 1983). Scholarship has expanded beyond residential contexts to activity-space and mobility perspectives, assessing structural segregation with unevenness in the distribution of groups across neighbourhoods (Ellis et al., 2004; Leetmaa et al., 2015) and exposure

segregation with realized or potential encounter among groups through everyday movements (Huang & Wong, 2015; Nilforoshan et al., 2023). Together, these approaches reveal patterns of similarity and separation in the spaces where people live, work, and circulate. Yet because studies adopt different socio-spatial assumptions and measurement choices, the “dimensions” under examination are often unclear and vary across places and datasets. For example, the specific “dimensions” analyzed differ across studies of income (Lichter et al., 2012), race (Cutler et al.,

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1999), gender (Qiao et al., 2025), aging (Lai et al., 2023), and are inconsistently linked to the mechanisms through which segregation manifests (Carter & Zimmerman, 2022; Glick et al., 2013; Kramer & Hogue, 2009; Williams & Collins, 2001). As a result, diagnoses for the same city can diverge depending on the framework employed. What, precisely, do we mean when we call a place “highly segregated,” and along which dimensions does that assessment rest?

We argue that progress requires an explicit dimensionalization of socio-spatial segregation that separates who differs socially from how segregation manifests spatially (Li et al., 2022). A conceptual framework that disentangles who is segregated (social dimensions) from how segregation materializes (spatial axes) is needed to resolve inconsistencies and to guide measurement. Accordingly, we propose a socio-spatial dimensional framework. The social dimensions are operationalized via the four Social Vulnerability Index (SVI) dimensions proposed by the Centers for Disease Control and Prevention and Agency for Toxic Substances and Disease Registry (CDC/ATSDR) (Flanagan et al., 2011; Ogunmayowa et al., 2025): socioeconomic status (SES), household composition and disability (HCD), minority status and language (MSL), and housing type (HT). The spatial axes comprise the assessment of metropolitan-level unevenness and community-level spatial clustering, which structure and shape how segregation is experienced. This framework yields a unified lens that reconciles structural and exposure traditions while adapting them to the mobility era.

To operationalize the framework, we utilize big mobility records from 45 million mobile devices in the U.S. and apply a mobility-based Social Segregation Index (SSI) to the 30 largest U.S. metropolitan statistical areas (MSAs). The SSI quantifies homophily in origin–destination (OD) flows, capturing MSA-scale exposure selectivity (Global SSI) and census tract-level exposure selectivity (Local SSI). We assess spatial clustering of Local SSI via Moran's I and identify compounded, multi-axis hot spots using Optimized Hot Spot Analysis (OHSA) with Jaccard overlap. We then highlight “extreme” tracts exhibiting four-axis compounding and examine their distribution across cities.

In sum, our study scales nationally and supports comparative inference across metropolitan areas, crucially indicating a tract that is highly segregated on one axis with strong clustering warrants different remedies than a tract that is moderately segregated on another axis without clustering. Thus, the proposed mobility-based socio-spatial framework enables the diagnosis of urban conditions and allows for the prioritization of dimension-specific interventions in social and urban management.

2. Literature review

2.1. How is social segregation defined?

Social segregation was initially grounded in early urban sociology, notably represented by the Chicago School. Park (1925) argued that segregation was a natural outcome of ecological differentiation, wherein social groups “naturally” inhabited homogeneous residential zones as part of urban spatial dynamics. Broadly speaking, two types of indices, the *dissimilarity* index and the *exposure* index, were majorly adopted, quantifying segregation from two distinct perspectives (Massey & Fischer, 1999).

The *dissimilarity* index was first proposed by Duncan and Duncan (1955), which centres on the spatial distribution/separation of different social groups. The core idea is to assess how unevenly these groups are distributed across neighborhoods, typically measured by calculating the proportion of a minority group needing to move to achieve a uniform spatial distribution. The *exposure* index instead conceptually focuses on the *experience* of segregation by measuring the likelihood of interaction between socially different groups within the same neighborhood.

Social segregation was first used to describe the residential environment. However, an increasing number of scholars have realized that social segregation is not a static phenomenon bound to neighbourhoods,

but a dynamic experience embedded in individuals' daily lives (Golledge, 1997; Sampson & Levy, 2020; Su et al., 2024). With the development of GPS techniques (Yang et al., 2024), personal mobile devices (Pan & He, 2024) and location-recording applications, personal activity spaces can now be more effectively quantified using digitally collected mobility traces. This advancement surpasses the limitations of conventional questionnaire-based methods and enables the examination of social segregation at much larger scales (Li & Wang, 2017). For example, Nilforoshan et al. (2023) examined segregation using 1.6 billion mobile phone mobility records to capture individuals' social exposure and found that segregation increased in large US cities. Xu et al. (2019) utilized 50 days of call detail records (CDR) from 2.1 million users in Singapore and performed a national-level examination of segregation across both communication space (phone-call-based) and physical space (mobility-based). Li et al. (2022) inherited Xu's framework and used mobility data collected from 45 million mobility devices to quantify the dynamics of social segregation in 12 most populated metropolitan areas in the US before and during the COVID-19 pandemic. As a result, the spatial scope of social segregation research has expanded from residential areas to activity spaces (Phillips et al., 2021; Wang et al., 2011). This expansion allows for a more comprehensive understanding of the times and places where individuals or groups encounter segregation, providing a basis for conducting social segregation research from a human mobility perspective (Li & Wang, 2017; Schnell & Yoav, 2001).

2.2. How are “dimensions” conceptualized and measured in segregation research?

Segregation is widely recognized as a multi-dimensional phenomenon that encompasses several distinct but interrelated aspects. In terms of different social dimensions, segregation quantifies the *similarity* or *dissimilarity* of residential spaces or interpersonal interactions among different social groups, such as concentrated poverty and segregation (Casey et al., 2017; Kohl et al., 2025; Quillian, 2012), physical conditions and disability (Gómez-Zúñiga et al., 2023; Lai et al., 2023), racial culture and languages (Glick et al., 2013; Stepinski & Dmowska, 2019; Su et al., 2024; Williams & Collins, 2001; Wu et al., 2023), housing and car ownership (Moya-Gómez et al., 2021; Park et al., 2021; Xu et al., 2022).

Besides that, a stream of segregation studies emphasized spatial organization of residential populations. Massey and Denton (1988) systematically articulated segregation as a multidimensional problem, identifying five core dimensions of residential segregation in spatial patterns: evenness, concentration, centralization, clustering, and exposure. In subsequent scholarship, the meaning of “multidimensionality” has gradually expanded beyond spatial form itself, increasingly referring to the coexistence of multiple *social* dimensions through which segregation operates and is experienced. The segregation of low-income households into distinct neighbourhoods creates “*poverty traps*” in space where disadvantages perpetuate across generations, which makes poverty into a concentration in space (Grove et al., 2018; Lichter et al., 2012). The segregation patterns can also be reflected at higher spatial scales. African Americans in more highly segregated cities had significantly worse economic outcomes, including higher unemployment and lower incomes than those in less segregated cities (Cutler et al., 1999).

Two main interpretations of “multi-dimensional indices” in segregation thus could be drawn from the interwoven studies. The first interpretation aligns with the concept of *structural* segregation, viewing segregation as a result of the multiple forms of discrimination or separation across various social-structural dimensions, such as education (Zhang et al., 2022), housing (Grove et al., 2018), and employment (Musterd, 2020). Some studies constructed separate indices for each dimension (Donegan & Tavares, 2024), for example, population, race, income, literacy, etc., while others have applied a composite index to represent overall segregation (Li et al., 2022; Xu et al., 2019). The

second interpretation centres on the *experience* of segregation, conceptualizing segregation as the accumulation of social differentiation through individuals' everyday encounters (Wu et al., 2023). Rooted in the people-based paradigm, this line of research examines segregation through patterns of individual mobility, exposure, and interaction behaviours, often incorporating multiple social attributes (e.g., income, race, housing status) or mechanisms (e.g., opportunity structures, social ties). In this context, multidimensionality refers to the coexistence and intersection of different exposure-based processes that shape an individual's socio-spatial experience. For example, F. Li and Wang (2017) analyzed dimensions such as age, income and house type in relation to individuals' daily activity spaces, emphasizing the segregation of exposure in everyday life. Ta (2023) proposed a four-dimensional framework—spatial form, opportunity exposure, spatiotemporal interaction, and social tie—for measuring activity space-based segregation.

2.3. Knowledge gaps and conceptualization of the dimensional framework

The knowledge gaps in existing studies are twofold. Theoretically, while many “dimensions” of segregation have been explored both in social composition and spatial form, the field lacks a shared schema for naming and measuring them consistently. This inconsistency creates challenges for cross-city and cross-dimensional comparisons, risks misdiagnosing the sources of inequality, and undermines targeted interventions. Methodologically, while big mobility data are increasingly used in segregation analysis, it remains unclear how to effectively construct multidimensional activity spaces with this data and examine their socio-spatial distributions. A clear understanding of how segregation manifests socio-spatially and where the most segregated areas are located remains lacking.

To address this gap, we define a dimension of socio-spatial segregation as the combination of social dimensions (who differs socially) and spatial axes (how segregation is structured and experienced). This two-part definition distinguishes composition from configuration, enabling measurement strategies to be mapped transparently to theory. We conceptualize segregation from a mobility-based perspective, which is operationalized by quantifying *exposure selectivity* during daily movements between OD pairs. This extends the classical notion of *exposure* from static *co-residence* to dynamic *co-presence* and *co-mobility* in urban space. To measure this, we adapt the “*unevenness*” and “*clustering*” axes from Massey and Denton's (1988) framework to quantify (1) the uneven spatial distribution of *exposure selectivity*, and (2) the extent to which areas of high *exposure selectivity* are continuous or spatially proximate. Furthermore, our multidimensional formulation incorporates multiple social dimensions. We explore how these attributes condition mobility opportunities and constraints, thereby resulting in segregation across different spatial axes. We operationalize the social dimensions using the four CDC/ATSDR SVI themes at the census-tract level: Socioeconomic Status (SES); Household Composition & Disability (HCD); Minority Status & Language (MSL); Housing Type (HT). These themes are widely used, comparable across U.S. metropolitan areas, and align with prior segregation scholarship that treats multidimensionality through either multiple social-structural separations or multiple attributes shaping exposure. By analyzing these four social dimensions against two spatial axes: the MSA-level unevenness and the tract-level clustering, we explore:

- **Coupling & Severity:** How does the severity of segregation vary across different social dimensions? To what extent do the coupling relations between social dimensions reinforce or become shaped by the overall severity of segregation?
- **Clustering & Severity:** Do socially segregated communities tend to be spatially clustered? Does spatial concentration (clustering) reinforce segregation severity, and how does this relationship differ across various social dimensions?

Table 1
Top 30 most populated MSAs in the U.S.

MSA Full Name (Short Name)	Population	% of the US population
New York-Newark-Jersey City (New York)	19,216,182	5.85
Los Angeles-Long Beach-Anaheim (Los Angeles)	13,214,799	4.03
Chicago-Naperville-Elgin (Chicago)	9,458,539	2.88
Dallas-Fort Worth-Arlington (Dallas)	7,573,136	2.31
Houston-The Woodlands-Sugar Land (Houston)	7,066,141	2.15
Washington-Arlington-Alexandria (Washington, DC)	6,280,487	1.91
Miami-Fort Lauderdale-Pompano Beach (Miami)	6,166,488	1.88
Philadelphia-Camden-Wilmington (Philadelphia)	6,102,434	1.86
Atlanta-Sandy Springs-Alpharetta (Atlanta)	6,020,364	1.83
Phoenix-Mesa-Chandler (Phoenix)	4,948,203	1.51
Boston-Cambridge-Newton (Boston)	4,873,019	1.48
San Francisco-Oakland-Berkeley (San Francisco)	4,731,803	1.44
Riverside-San Bernardino-Ontario (Riverside)	4,650,631	1.42
Detroit-Warren-Dearborn (Detroit)	4,319,629	1.32
Seattle-Tacoma-Bellevue (Seattle)	3,979,845	1.21
Minneapolis-St. Paul-Bloomington (Minneapolis)	3,640,043	1.11
San Diego-Chula Vista-Carlsbad (San Diego)	3,338,330	1.02
Tampa-St. Petersburg-Clearwater (Tampa)	3,194,831	0.97
Denver-Aurora-Lakewood (Denver)	2,967,239	0.90
St. Louis (St. Louis)	2,803,228	0.85
Baltimore-Columbia-Towson (Baltimore)	2,800,053	0.85
Charlotte-Concord-Gastonia (Charlotte)	2,636,883	0.80
Orlando-Kissimmee-Sanford (Orlando)	2,608,147	0.79
San Antonio-New Braunfels (San Antonio)	2,550,960	0.78
Portland-Vancouver-Hillsboro (Portland)	2,492,412	0.76
Sacramento-Roseville-Folsom (Sacramento)	2,363,730	0.72
Pittsburgh (Pittsburgh)	2,317,600	0.71
Las Vegas-Henderson-Paradise (Las Vegas)	2,266,715	0.69
Austin-Round Rock-Georgetown (Austin)	2,227,083	0.68
Cincinnati (Cincinnati)	2,221,208	0.68
Total	149,030,162	45.40

- **Convergence & Independence:** To what extent are the spatial patterns of different social dimensions coupled? We assess areas of convergence (where multiple forms of severe segregation manifest in the same space) versus independence (where they manifest in distinct locales).

This framework integrates classic structural notions of segregation with contemporary mobility-era concerns regarding contact and clustering within activity spaces constructed by massive human mobility patterns. Leveraging large-scale mobility data from the 30 most populated U.S. MSAs and widely adopted SVI variables, our analysis examines national comparability, utilizes the OD-based SSI for exposure, Moran's I/OHSA for clustering, and severity tertiles for regime classification, providing a replicable blueprint for metropolitan-scale comparisons.

3. Study areas and data

3.1. Study areas

In this paper, we selected the 30 MSAs with the largest populations in the U.S., as shown in Table 1. The data were obtained from the U.S.

Census Bureau.¹ Table 1 presents the population figures and the percentage of the total U.S. population for each MSA. Approximately 45.4% of the U.S. population resided in the examined MSAs. The spatial distribution of the selected MSAs is illustrated in Fig. 1, labelled with the short names defined in Table 1.

3.2. Data sources

3.2.1. CDC/ATSDR SVI

The Social Vulnerability Index (SVI), developed by the CDC/ATSDR, is a national annual survey that has been extensively utilized in environmental science, urban studies, and social sciences over the past decade (Flanagan et al., 2011). The CDC/ATSDR SVI includes 15 selected variables representing different aspects of social vulnerability, which are grouped into four thematic dimensions: Socioeconomic Status (SES), Household Composition and Disability (HCD), Minority Status and Language (MSL), and Housing Type (HT), as detailed in Table A1. The SES quantifies a community's access to financial resources through variables such as poverty, unemployment, income, and educational attainment. The HCD identifies vulnerable demographic groups, including the proportion of elderly and young residents, individuals with disabilities, and single-parent households. The MSL highlights communities with a higher percentage of minority populations and residents with limited English proficiency. Finally, the HT captures structural and household-level vulnerabilities, such as crowded housing, mobile homes, and households without access to a vehicle. Although the latter may also have implications for accessibility, in this study it is treated as a structural social vulnerability indicator within the HT dimension to adhere to SVI's original structure for easy replication.

These four sub-themes closely align with the social dimensions examined in prior studies, as presented in our literature review, making the SVI a suitable data source for capturing multi-dimensional social structural characteristics (Ogunmayowa et al., 2025).

3.2.2. Mobility data

The mobility data used in this study were obtained from SafeGraph (<https://www.safegraph.com/>), a commercial provider that aggregates anonymized location information collected from a diverse set of mobile applications. These data offer valuable insights into population movements and visitation patterns to physical locations across the United States. It is worth noting that SafeGraph has since merged with Dewey (<https://www.deweydata.io/>), where its foot-traffic datasets are now hosted.

Specifically, we utilize SafeGraph's publicly available *Social Distancing Metrics* dataset (*Social Distancing Metrics | SafeGraph Docs*, 2022), which provides daily mobility information at the Census Block Group (CBG) level for the entire conterminous United States throughout 2019 (January 1–December 31). The data are derived from GPS signals of approximately 45 million anonymized mobile devices, representing roughly 10% of all devices in the United States (Huang et al., 2022), providing a sufficiently large and representative sample for analyzing human spatial interactions. SafeGraph determines each device's home location based on frequent nighttime presence over a six-week observation window, assigning it to a Geohash-7 grid (~153 m × 153 m). Aggregated daily movements from these inferred home locations to CBG-level destinations are then recorded, enabling the construction of origin–destination matrices that characterize typical daily mobility and travel patterns (Li et al., 2021). The Social Distancing Metrics dataset has been widely recognized as a valuable mobility resource for examining how people navigate and interact across urban and rural environments (Hu et al., 2021; Huang et al., 2022).

4. Methods

Fig. 2 illustrates the workflow of this study.

First, we adopted the method for calculating the Social Segregation Index (SSI) as described by Li et al. (2022), and computed the SSI for the 30 most populated MSAs in the U.S. along four distinct social dimensions: socioeconomic status (SES), household composition and disability (HCD), minority status and language (MSL), and housing type (HT), as well as a composite social vulnerability index (Comp.) for comparison. We computed the SSI at two spatial axes: the MSA-level (Global SSI) and the census tract-level (Local SSI). Next, we statistically examined the correlations between Global SSI scores across different social dimensions to explore how segregation manifests across them. We then investigated the spatial distribution of Local SSI values to determine whether highly segregated communities (census tracts) tend to cluster within highly segregated MSAs and how this clustering varies by social dimensions. Moran's I statistic was used to quantify the spatial autocorrelation of Local SSI values at the census tract level, which we further compared with the Global SSI through correlation analysis. Finally, we identified and mapped extremely segregated communities facing segregation across multiple dimensions by performing Optimized Hot Spot Analysis (OHSA) and spatial overlay analysis. This allowed us to identify and combine “hot spots” of highly segregated census tracts across the various dimensions. Additionally, we conducted the Jaccard Similarity Test to quantify the spatial similarity of identified “hot spots” among the different dimensions.

4.1. Global and local SSI calculation

In this study, we adapted the method used in Xu et al. (2019) to quantify social segregation. This method is based on the concept of mobility homophily—the tendency for people to travel to places that are socially similar to where they reside. If residents of a given neighborhood primarily visit other “socially similar” neighbourhoods, this pattern is considered a form of segregation.

Census tracts were chosen as the basic unit of analysis due to their consistent population size, the availability of sociodemographic data, and their widespread use in representing neighbourhoods or communities in social science research. Interactions between census tracts are represented by origin (O) and destination (D) mobility pairs. Two measures were calculated:

- Global SSI: a measure quantifying the social similarity of all OD pairs within an MSA;
- Local SSI: a measure quantifying the social similarity of all OD pairs originating from a single census tract.

To assess how “socially similar” OD pairs are, we first ranked all census tracts within each MSA based on their value in a selected social dimension, referring to four distinct themes (e.g., SES, HCD, MSL, and HT) as well as the composite social vulnerability index from CDC SVI in this study. This ranking process produced a sequence of social ranks, $(Rank_i)_{i=1}^n = \{1, 2, 3, \dots, n\}$, where $Rank_i$ represents the rank of the i -th tract out of n tracts in the MSA.

For each OD pair, for example between origin tract a and destination tract b , we calculated the absolute difference in their social ranks, $|Rank_a - Rank_b|$, and compared it to the absolute differences between tract a to all other tracts within the MSA, denoted as $|Rank_a - Rank_{other}|$, where $Rank_{other} \in (Rank_i)_{i=1}^n$, $other \neq a, b$. By doing so, we can obtain two sets of tracts:

- Set $A = \{other \mid |Rank_a - Rank_{other}| < |Rank_a - Rank_b|\}$, consisting of tracts that are closer in social rank to tract a than tract b is.
- Set $B = \{other \mid |Rank_a - Rank_{other}| = |Rank_a - Rank_b|\}$, consisting of tracts that are equidistant in social rank to tract a as tract b is.

¹ Resource: <https://www2.census.gov/programs-surveys/popest/tables/2010-2019/metro/totals/cbsa-met-est2019-annres.xlsx>

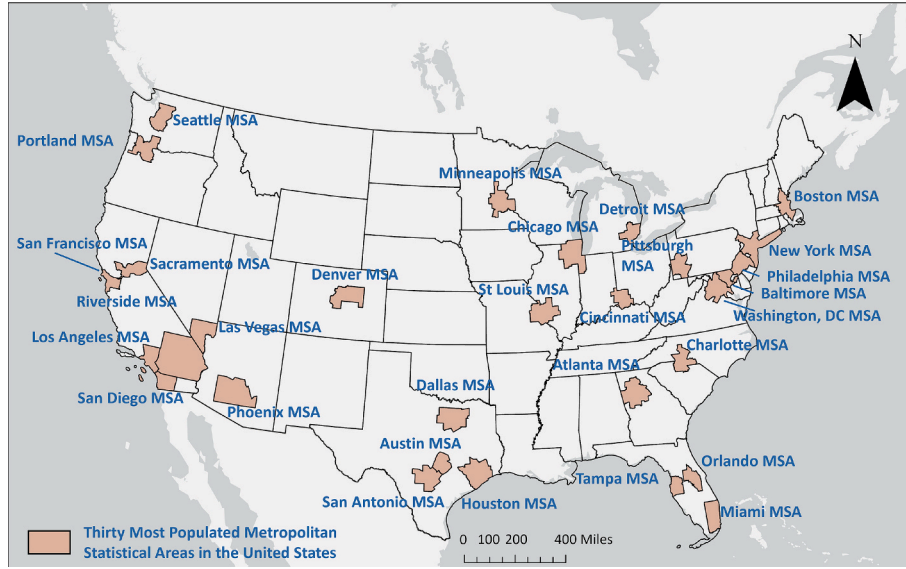


Fig. 1. The spatial distribution of the top 30 most populated MSAs in the U.S.

The social distance from tract a to tract b , denoted as $SD_{a \rightarrow b}$, was then calculated as:

$$SD_{a \rightarrow b} = \begin{cases} \frac{|A|}{n-1}, & \text{if } B \text{ is } \emptyset \\ \frac{|A| + 0.5|B|}{n-1}, & \text{otherwise} \end{cases}, \quad (1)$$

where $|A|$ and $|B|$ represent the cardinalities of set A and B , respectively.

The social distance represents the proportion of tracts that are closer in social rank to tract a than tract b is, with values ranging from 0 to 1. A higher value indicates that the two tracts are more “dissimilar” in their social ranks. To ease the result interpretation, where higher values represent greater social similarity, we calculate the social similarity as $SS_{a \rightarrow b} = 1 - SD_{a \rightarrow b}$, which also ranges from 0 to 1, with higher values indicating higher similarity.

4.1.1. Global SSI

This study calculated the Global SSI as the weighted average of social similarity across all OD pairs along a selected social dimension within a given MSA. We first grouped all mobility records into a set of unique OD pairs $\{OD_1, OD_2, OD_3, \dots, OD_m\}$, where m denotes the total number of unique OD pairs. For each OD pair OD_p , we counted the number of associated mobility records, generating a set $\{Count_1, Count_2, Count_3, \dots, Count_m\}$, where $Count_p$ denotes the number of records for OD_p . As all mobility records have been aggregated to the census tract level, as mentioned above, the social similarity for each OD pair can be calculated using the previously defined methods. The Global SSI is then computed as:

$$Global\ SSI = \frac{\sum_{p=1}^m Count_p \bullet SS_{OD_p}}{\sum_{p=1}^m Count_p} \quad (2)$$

where SS_{OD_p} denotes the social similarity for the p -th OD pair.

4.1.2. Local SSI

Similarly, this study calculated a Local SSI as the weighted average of social similarity for all mobility records originating from the same census along a selected social dimension. For a given tract s , we first extracted a subset of distinct OD pairs that originate from a tract s , denoted as $OD_{p|s}$. Correspondingly, we subset the count of mobility records associated with these OD pairs, denoted as $Count_{p|s}$. Then, the

Local SSI for census tract s can be calculated as follows:

$$Local\ SSI(s) = \frac{\sum_{p=1}^o Count_{p|s} \bullet SS_{OD_{p|s}}}{\sum_{p=1}^o Count_{p|s}}, \quad (3)$$

where o represents the total number of distinct OD pairs originating from tract s , and $p|s$ indicates the p -th OD pair with origin tract s .

As demonstrated by Xu et al. (2019), the use of relative rank differences—rather than an absolute measure—to represent social distance establishes a consistent theoretical baseline of 0.5. This value indicates that individuals interact randomly with others in social or physical space. In contrast, calculations based on absolute rank differences yield varying baselines depending on an individual's rank, which complicates meaningful comparisons across individuals or cities. Operationally, this means that if residents from a census tract travel equally to all other tracts, the Local SSI would be 0.5. A value closer to 1 indicates that the tract tends to interact primarily with socially similar tracts, while a value closer to 0 suggests more frequent interaction with socially dissimilar tracts. Similarly, the Global SSI also ranges from 0 to 1, with values approaching 1 indicating a higher degree of segregation—where residents in the MSA predominantly travel to socially similar tracts along the selected social dimension.

4.2. Statistical correlation analysis

To assess how segregation manifests along different dimensions, we utilized correlation analysis among the Global SSI values obtained from the four selected social dimensions across the 30 MSAs. We also compared them with the Global SSI calculated using a composite social vulnerability. To choose the appropriate correlation analysis method (Pearson's or Spearman's rank), we first applied the Shapiro-Wilk test to examine whether the distribution of the Global SSI values along different dimensions follows a normal distribution, which will inform the selection of correlation methods: Pearson's correlation was used for normally distributed Global SSI values, while Spearman's rank correlation was applied for non-normally distributed ones.

In addition to performing correlation analysis across all 30 MSAs, we explored whether—and how—the relationship between different dimensions of segregation varies among MSAs with differing segregation levels. To achieve this, we further conducted a sliding-window correlation analysis. First, MSAs were ordered based on their Global SSI values for the Composite dimension (Global SSI-Comp.). A fixed-size sliding window (window size = 15 MSAs; step = 1) was then moved

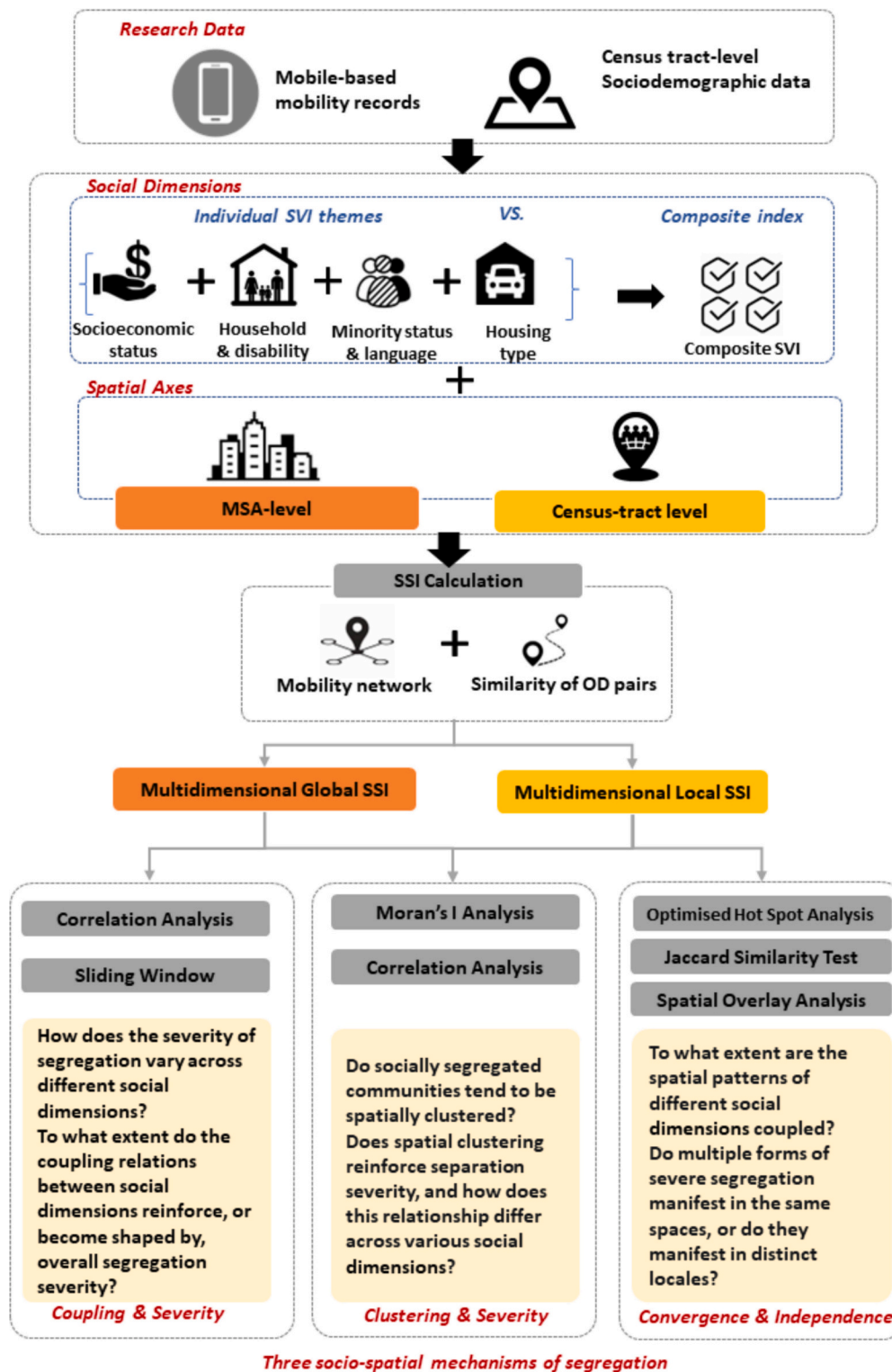


Fig. 2. Research workflow.

along this ordered sequence from low to high Global SSI-Comp. We selected a window size of 15 samples for two reasons: (1) prior research suggests that Spearman's rank correlation becomes unstable with fewer than 10–12 observations, while estimates based on 15 or more samples show substantially improved robustness (Conover, 1999; Gibbons & Chakraborti, 2011), (2) given the relatively small number of MSAs, a window size of 15 balances local sensitivity with statistical stability.

4.3. Spatial autocorrelation analysis

To explore whether socially segregated communities tend to be spatially clustered—and how this pattern differs across social dimensions—we calculated Moran's I statistic for local SSI values across different dimensions within each MSA. Moran's I statistic is a widely used measure of spatial autocorrelation, quantifying the degree to which similar attribute values are geographically clustered (Moran, 1950). The

statistic ranges from approximately -1 to $+1$: positive values indicate spatial clustering of similar values, negative values suggest spatial dispersion or dissimilarity, and values near zero imply spatial randomness (Anselin, 1995).

Additionally, to examine whether spatial clustering is more likely to occur in highly segregated areas, we analyzed the correlations between Global SSI scores and Moran's I values across 30 MSAs for different social dimensions.

4.4. Extremely segregated census tracts detection

To further identify census tracts exhibiting high levels of segregation across multiple social dimensions—referred to in this study as *extremely segregated census tracts*—we implemented Optimized Hot Spot Analysis (OHSA). This method was used to detect hot spots, defined as highly segregated census tracts that are spatially surrounded by other similarly segregated tracts, for each individual social dimension. For each dimension, tracts identified as hot spots were coded as 1, and all others as 0. We then combined these binary layers through spatial overlay analysis to generate a composite score for each tract, indicating the number of dimensions (ranging from 0 to 4) in which it was identified as a hot spot.

OHSA is a robust spatial technique for identifying statistically significant clusters. It integrates the widely used Getis-Ord G_i^* (G_i^*) statistic with an incremental search process to determine the optimal spatial scale for analysis. Rather than relying on a manually specified distance for computing the G_i^* statistic, OHSA calculates the Global Moran's I statistic across a series of increasing distances and automatically selects the one that yields the highest Moran's I value, representing the strongest spatial clustering tendency. The G_i^* statistic is then calculated using this optimal distance to identify hot spots, or clusters of high values. In this study, hot spots are identified based on both G_i^* statistic and the examined Local SSI values. Only those census tracts that (1) are statistically significant clusters (i.e., $p < 0.01$, $z > 2.58$) and (2) have a Local SSI value in the top tercile (i.e., the highest 33%) of all examined tracts are considered valid hot spots. This ensures that the detected hot spots are not only surrounded by similarly high values in space but also exhibit substantively high levels of social segregation themselves.

In addition, we used the Jaccard similarity coefficient to quantify the degree of spatial overlap among the hot spots identified across the four social dimensions. The Jaccard similarity coefficient is defined as:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (4)$$

where A and B are binary classifications representing identified hot spots in two different dimensions; $|A \cap B|$ represents the number of the census tracts identified as hot spots in both dimensions; $|A \cup B|$ represents the number of tracts identified as hot spots in either dimension.

The coefficient ranges from 0 to 1, where 0 indicates no overlap between the spatial patterns and 1 indicates perfect spatial correspondence. We calculated the Jaccard similarity coefficients for each pair of social dimensions, using only those hot spots identified at the 95% significance level, for each city in the study. This analytical approach offers valuable insights into the spatial alignment of different dimensions of segregation, the consistency of clustering patterns across social indicators, and the potential compounded effects of multiple forms of segregation in specific neighborhoods.

5. Results

5.1. The variation of segregation severity along different social dimensions

We calculated the Global SSI values for all 30 MSAs across the four selected social dimensions, as well as the composite social vulnerability

dimension for comparison, as shown in Fig. 3. The violin plots demonstrate that the severity of segregation manifests differently across the social dimensions. In particular, the social dimensions of SES and minority status and language (MSL) show noticeably higher Global SSI values than household composition and disability (HCD) and housing type (HT) among the examined MSAs. Such variations could easily be overlooked if relying solely on the composite social vulnerability index (Comp.).

In addition, we ranked the 30 MSAs based on their Global SSI values across the selected social dimensions, from highest to lowest. A higher rank indicates more severe segregation along a given dimension. As shown in Fig. 4, the rankings of MSAs vary considerably across different dimensions. For example, based on the composite social vulnerability (Comp.) dimension, New York is the most segregated MSA. When breaking down into different social dimensions, it shows high levels of segregation in housing type (HT, ranked first), socioeconomic status (SES, ranked fourth), and minority status and language (MSL, ranked fourth). However, its ranking for household composition and disability (HCD) is mid-range, at 17th out of the 30 MSAs. Detroit, which ranks second in the Comp. dimension, exhibits the highest levels of segregation in both SES and HCD, but ranks much lower in MSL (18th) and HT (23rd). Miami, on the other hand, is ranked mid-range overall (15th in the Comp. dimension) and relatively low in SES (18th) and HCD (26th), yet it displays the highest segregation in the MSL dimension and is highly ranked in HT (6th). These results further illustrate the multi-dimensional nature of segregation across the studied regions. Different MSAs face predominant segregation challenges along specific social dimensions.

5.2. Statistical correlations of segregation severity across social dimensions

To assess the correlations of segregation severity across different social dimensions, we first applied the Shapiro-Wilk test to the Global SSIs of the 30 MSAs for each dimension. While most social dimensions passed the normality test (p -value > 0.05), the HT dimension returned a p -value of 0.0331, indicating a significant deviation from normality. Therefore, we employed Spearman's rank correlation coefficient (r) to account for the presence of non-normally distributed data.

As shown in Fig. 5, the Global SSI values obtained across the three selected social dimensions (HCD, MSL, HT) exhibited statistically weak ($0.2 < r \leq 0.4$) to moderate ($0.4 < r \leq 0.6$) correlations with the Global SSI values for the Composite dimension (Global SSI-Comp.). Only Global SSI-SES demonstrated a very strong correlation ($0.8 < r \leq 1$) with Global SSI-Comp. These results further underscore the necessity of examining social segregation across different social dimensions, as using a composite SVI alone cannot capture the nuanced multi-dimensional manifestations of social segregation. Among the four selected dimensions, SES and HCD showed a moderate correlation ($r = 0.53$), while MSL and HT also exhibited a moderate correlation ($r = 0.47$). These results indicate that, at the MSA level, socioeconomic segregation and household composition segregation are more likely to be coupled. Similarly, minority segregation appears to be more closely associated with housing and transportation segregation.

We further assessed how correlations between different pairs of social dimensions varied along the Global SSI-Comp. gradient. Fig. 6 presents the sliding-window Spearman correlation results for each pair. The x-axis represents the mean Global SSI-Comp. value among the 15 MSAs in each window, ordered from low to high segregation. The y-axis shows the Spearman's rank correlation coefficient (ρ).

When compared with the Comp. dimension, only the SES dimension had a strong and consistent correlation ($0.6 < \rho \leq 0.8$) with Comp., further underscoring the importance of analyzing segregation along disaggregated dimensions. HT and Comp. also exhibited a moderate positive association among MSAs with lower segregation levels. Among these less segregated MSAs, we also observed a moderate positive

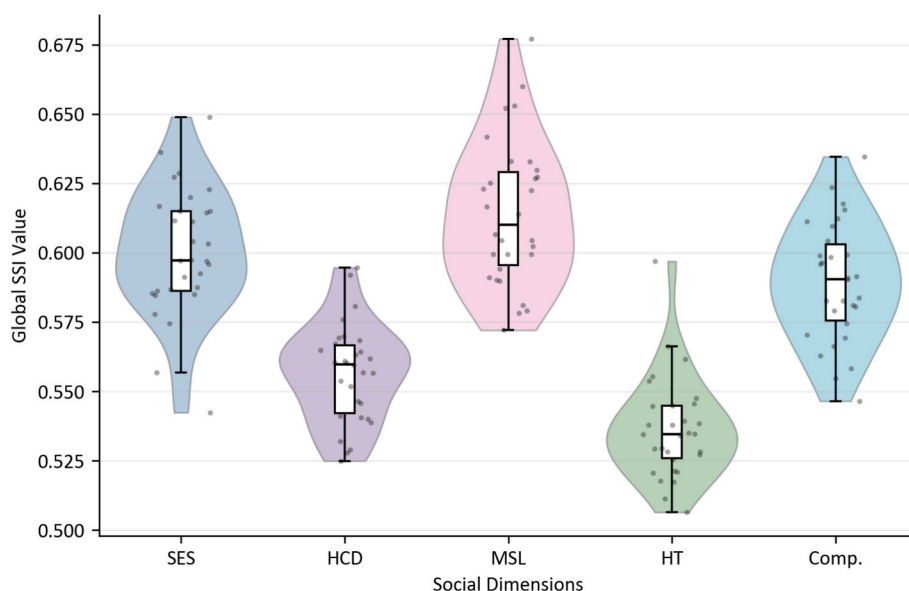


Fig. 3. Violin plots showing the distribution of Global Social Segregation Index (SSI) values across four social dimensions and the composite index for 30 Metropolitan Statistical Areas (MSAs). The dimensions are: Socioeconomic Status (SES), Household Composition & Disability (HCD), Minority Status & Language (MSL), Housing Type (HT), and the overall Composite Index (Comp.).

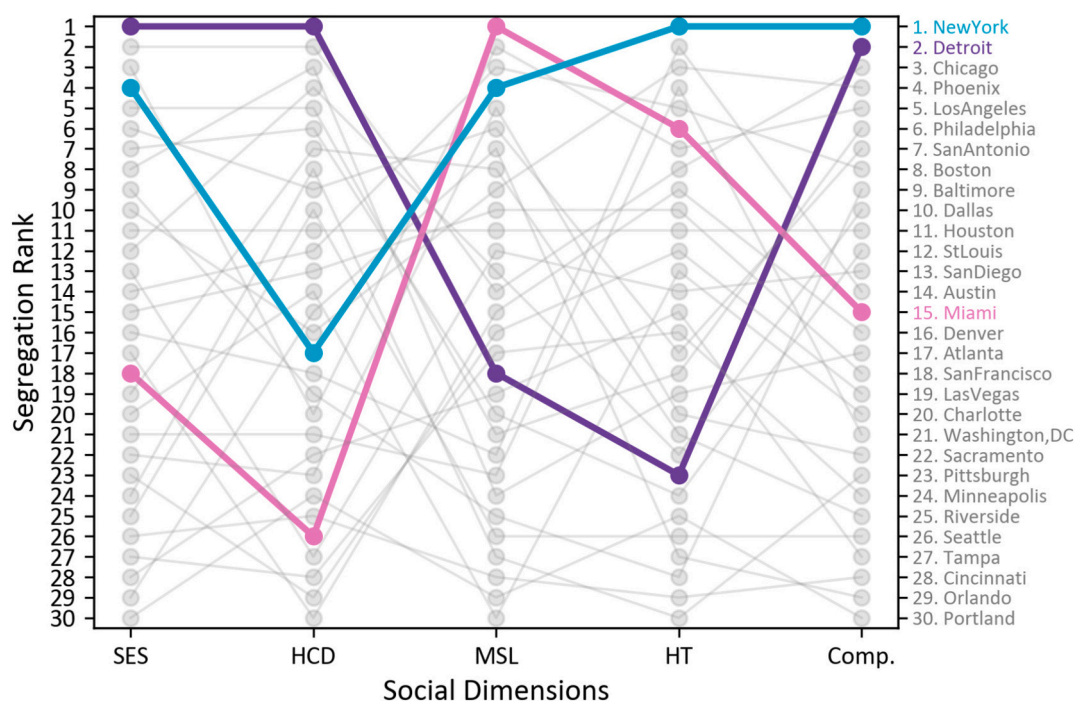


Fig. 4. Segregation rankings of 30 Metropolitan Statistical Areas (MSAs) across different social dimensions. Highlighted lines indicate the highest-ranking MSAs for each dimension.

correlation between SES and HT, suggesting a potential coupling mechanism between these two dimensions. As segregation levels increased to the mid- and high-Global SSI-Comp. range, SES and HCD became the most strongly correlated pair. Although some windows showed a significant negative correlation between HT and SES, this pattern was inconsistent and warrants further investigation.

5.3. Spatial distribution of segregated neighbourhoods within each MSA

To explore the spatial clustering tendency of highly segregated neighborhoods (census tracts), we calculated Moran's I statistics for all

30 MSAs based on their Local SSI values derived from each social dimension, as illustrated in Fig. 7. Overall, all MSAs exhibited statistically significant (p -values < 0.001), mild to moderate spatial clustering across the four selected social dimensions and the comparison (Comp.) dimension. Among them, segregation along the MSL dimension showed the highest spatial clustering tendency, with a mean Moran's I value of 0.483 across the 30 MSAs, followed by SES (0.457). The HCD (0.326) and HT (0.268) dimensions exhibited lower levels of spatial clustering compared to the Comp. dimension (0.425).

Among all examined MSAs, Los Angeles showed the highest spatial clustering tendency, ranking 1st in SES, MSL, and Comp., and 3rd in HT,

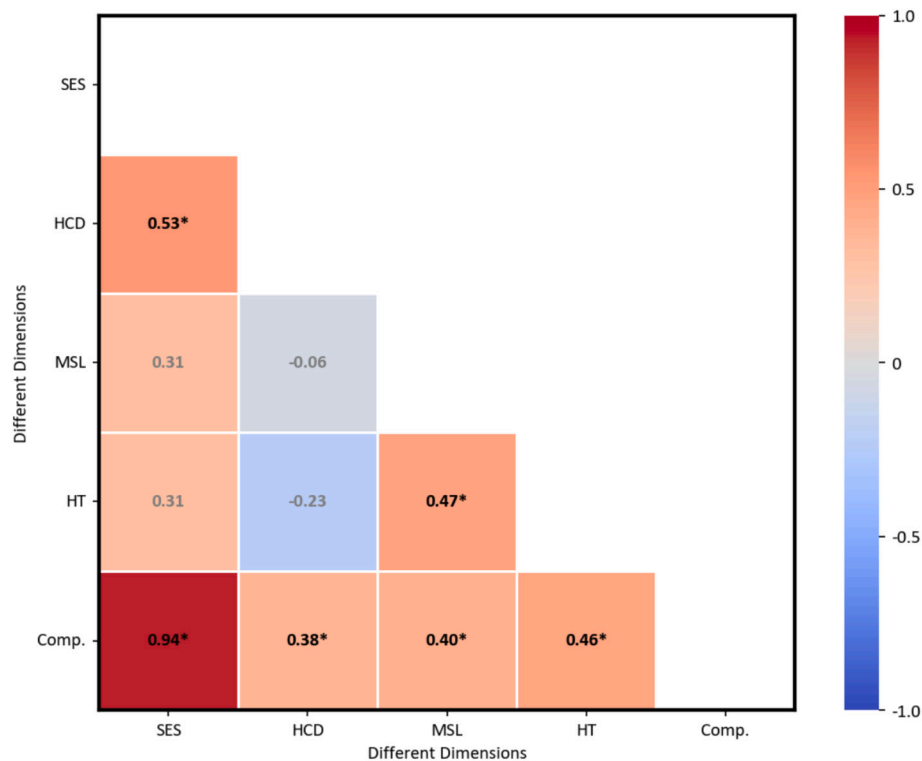


Fig. 5. Correlation matrix of multi-dimensional Global SSI values for all 30 MSAs.

suggesting that socially segregated neighbourhoods tend to be spatially clustered in Los Angeles. Sacramento ranked highest in HCD, while Washington, D.C., ranked highest in HT.

We further performed Spearman's correlation analysis between Moran's I index and the Global SSI values calculated across different social dimensions. The results are presented in Table 2. Three of the selected social dimensions—SES, MSL, and HT—as well as the Comp. dimension, showed moderate positive correlations ($r \approx 0.5$) between spatial clustering (Moran's I) and overall segregation severity (Global SSI). Only the HCD dimension did not show a statistically significant correlation. These results suggest a trend in which MSAs with higher segregation levels along SES, MSL, and HT dimensions tend to exhibit stronger spatial clustering of highly segregated neighborhoods (census tracts). However, this pattern was not observed for the HCD dimension in the examined MSAs.

5.4. Extremely segregated census tracts detection

5.4.1. Hot spots identification and Jaccard similarity assessment

To further identify census tracts facing compounded segregation across multiple social dimensions, we identified the hot spot census tracts from each social dimension separately using OHSA, which were then compared and combined using the Jaccard similarity coefficient and spatial overlay analysis, respectively. To ensure the identified hot spots exhibit both high segregation values and high spatial clustering, they are defined as those with OHSA confidence levels at 99% and above (i.e., $p < 0.01$, $z > 2.58$) and Local SSI values in the top tercile of the MSA.

For example, Fig. 8 illustrates the hot spot census tracts from the New York MSA across four selected dimensions. A total of 505 census tracts are identified as hot spots in the SES dimension, followed by 585 in HCD, 560 in MSL, and 814 in HT.

We further quantified the similarity of hot spot tracts identified across different social dimensions for each MSA using the Jaccard similarity index. Fig. 9 shows the results for the New York MSA, indicating substantial variation in hotspot similarity across dimensions. The

highest similarity is observed between MSL and SES, with a value of 0.41, while similarities among other dimension pairs remain below 0.3.

We applied the same analysis to all other MSAs and found that the similarity of hot spots between different social dimensions is generally low. The highest value observed was 0.6, between SES and MSL in the Phoenix MSA, as shown in Fig. A1. These results further demonstrate the spatial variation in highly segregated census tracts across different social dimensions.

5.4.2. Extreme cases identification

By overlapping the results from all four dimensions, we identified census tracts with compounded segregation issues. Fig. 10 illustrates the results for the New York MSA. Each census tract is assigned a value from 0 to 4, indicating the number of social dimensions in which it was identified as a hot spot.

The spatial distribution of extremely segregated tracts (score = 4) is illustrated in Fig. 10. A total of 3.33% of census tracts received a score of 4, and these are primarily concentrated in three regions. Region A is the well-known Bronx area, long recognized as a site of systemic challenges shaped by historical forces such as freeway and expressway construction, redlining, and the Bronx fires. This area remains a powerful example of entrenched contradictions and urban conflict. Region B (Wantagh, Seaford, Bellmore) exemplifies a historically exclusive, affluent white suburb shaped by postwar zoning policies, redlining, and racial steering. Despite its proximity to highly segregated minority neighborhoods, strict zoning regulations and fragmented school districts have maintained a sharp spatial and racial divide. Region C (Northport) represents a demographically older, high-income white community characterized by low diversity and limited transit connectivity. Its segregation is shaped by its relatively remote location and preservation-oriented development. Structural exclusion persists through a restrictive housing supply and socioeconomic homogeneity.

We further calculated the extremely segregated census tract detection results (score = 4) for all 30 MSAs. Among them, only six MSAs had any extremely segregated tracts detected: New York (3.33% of census tracts), Philadelphia (1.92%), Los Angeles (0.48%), Detroit (0.39%),

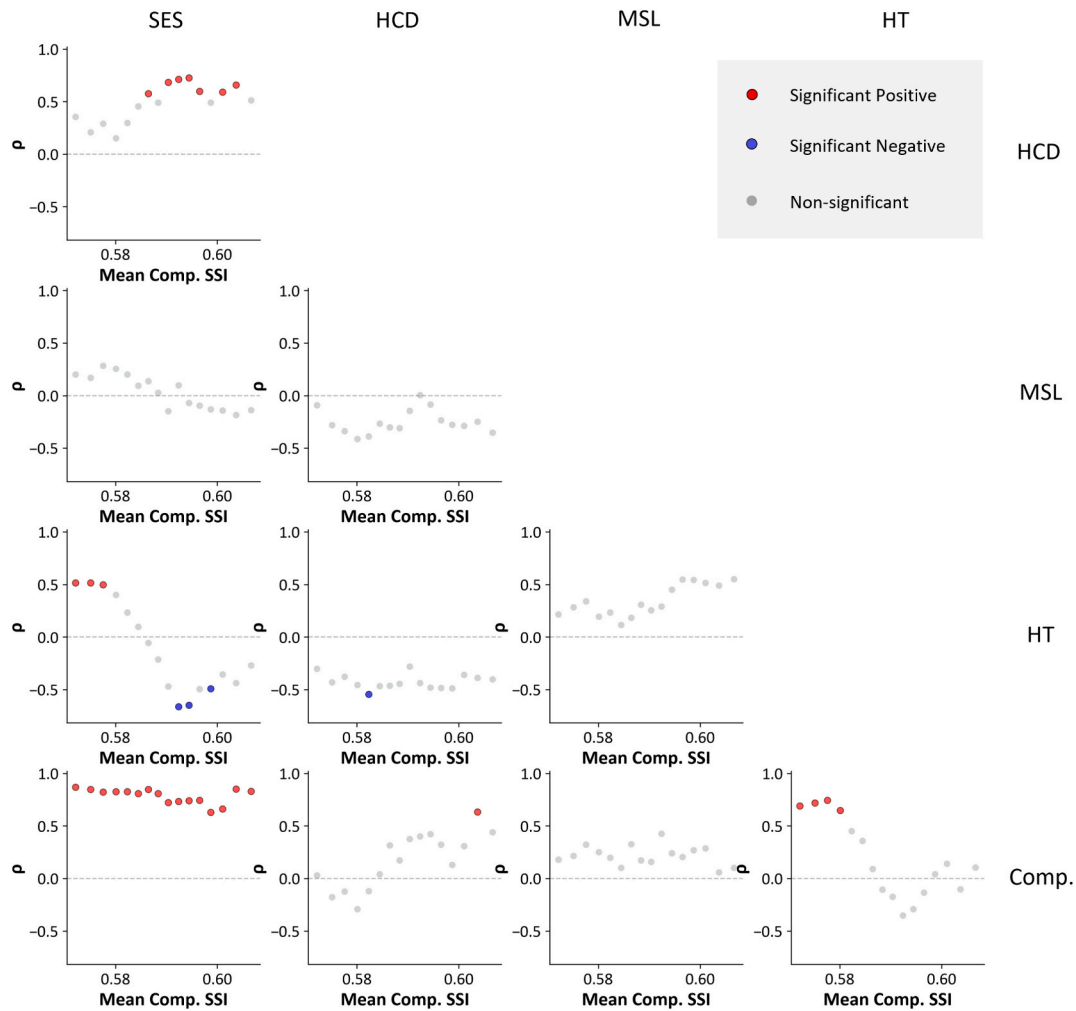


Fig. 6. Sliding-window correlation results. Red points represent significant positive correlations, blue points indicate significant negative correlations, and gray points denote non-significant correlations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Portland (0.20%), and Dallas (0.076%). These results show that New York and Philadelphia have the highest proportion of census tracts facing compounded segregation issues among the examined MSAs. We also document the number and proportion of census tracts identified as hotspots from at least one dimension (score > 0) for each MSA, as detailed in Table A2. From this perspective, Los Angeles has the highest ratio of hotspot tracts identified from at least one dimension (23.35%), followed by Las Vegas (23.20%), and San Francisco (22.22%). Although New York ranks fifth in terms of proportion (18.75%), it has the highest absolute number of hotspot census tracts, with a total of 838. These results indicate that these MSAs may experience predominant segregation issues along specific social dimensions.

6. Discussion

6.1. Key findings

By using the proposed socio-spatial dimensional framework across the 30 largest U.S. MSAs, this study asked a deceptively simple but essential question: *what do we mean when we call a place “highly segregated” and along which dimensions does that judgment rest?* Our findings show that the meaning of “high segregation” depends on the axis observed and the mechanism captured. Global SSI reveals systematic differences by social dimensions, with SES and MSL generally exhibiting higher metro-scale exposure selectivity than HCD or HT, while Local SSI, Moran's I, and OHSA illuminate the mechanisms that can amplify or

dampen the lived experience of segregation in neighborhoods. This reinforces longstanding arguments that segregation is multi-dimensional and that different measures tap different underlying processes, from structural unevenness to exposure and clustering (Cai et al., 2024; Clark, 1986; Loo et al., 2024; Massey & Denton, 1988; Reardon & O'Sullivan, 2004).

6.2. Link to existing literature

Across the 30 MSAs, the distribution of Global SSI varies markedly by axis; SES and MSL tend to show higher severity than HCD and HT. Moreover, rankings by axis reorder cities in substantively important ways: New York is top-ranked on the composite, but its axis-specific profile is first on HT, fourth on SES and MSL, and only mid-range on HCD; Detroit leads on SES and HCD but ranks low on MSL and HT; Miami is mid-range on the composite yet tops MSL and ranks high on HT. This pattern echoes prior findings that race/ethnicity and socioeconomic status often dominate MSA-level segregation, while other dimensions such as disability and transport limitations are more context-dependent (Loo et al., 2024; Moya-Gómez et al., 2021; Park et al., 2021; Qiao & Yeh, 2023). The re-orderings also caution against reliance on composite scores that can mask dimension-specific challenges (Li & Wang, 2017; Mahdavi Ardestani et al., 2018; Park & Kwan, 2018; Ta, 2023; Zhang et al., 2021). These axis-specific re-orderings are not merely methodological artifacts but reflect fundamentally different inequality-generating mechanisms. Prior research shows that

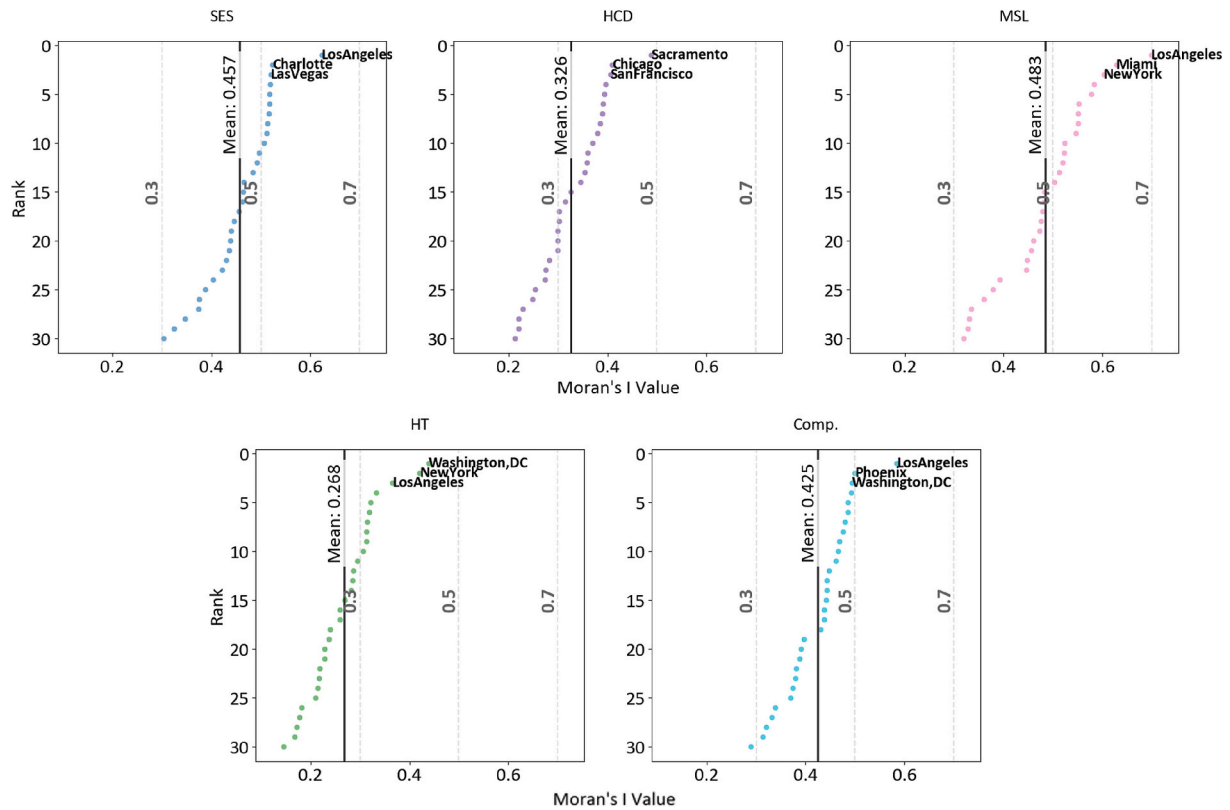


Fig. 7. Moran's I statistics for 30 MSAs across different social dimensions. (All values are statistically significant at p -values < 0.001).

Table 2

Correlation coefficients between Moran's I index and Global SSI value along different social dimensions.

Social Dimensions	Correlation coefficients	p-Value
SES	0.45	0.01*
HCD	0.31	0.09
MSL	0.55	0.001*
HT	0.54	0.002*
Comp.	0.52	0.002*

segregation along socioeconomic and racial/linguistic axes is strongly associated with unequal access to quality housing, employment opportunities, and neighborhood resources, whereas segregation related to disability, household structure, or transport constraints often manifests through barriers to daily mobility, service access, and social participation (Carter & Zimmerman, 2022; Casey et al., 2017; Park et al., 2021; Williams & Collins, 2001).

In terms of Local SSI, the results highlight that clustering matters. Local SSI surfaces neighborhood-level exposure selectivity, and its spatial organization differs by axis. MSAs show mild-to-moderate clustering overall, but MSL exhibits the highest clustering on average, followed by SES, with weaker clustering for HCD and HT. Moreover, clustering increases with severity for SES, MSL, and HT (moderate positive correlations between Moran's I and Global SSI), but not for HCD, which indicates that family/disability structures may raise exposure selectivity without forming tight spatial clusters. This axis-specific differentiation aligns with mobility-based exposure work showing that contact opportunities vary by group (Nilforoshan et al., 2023; Phillips et al., 2021; Sampson & Levy, 2020; Xu et al., 2019), and with studies emphasizing that disability-related isolation is often socially structured but not tightly spatially concentrated (Gómez-Zúñiga et al., 2023; Lai et al., 2023). This underscores why exposures and clustering should be distinguished analytically. From an equity perspective, spatial clustering

implies that exposure-based segregation is not randomly distributed but concentrated in specific neighborhoods, where disadvantages related to housing quality, transport accessibility, and environmental risks can accumulate and reinforce one another. This echoes findings that clustered segregation amplifies unequal life chances by shaping daily activity spaces, limiting access to opportunity-rich areas, and constraining mobility choices (Nilforoshan et al., 2023; Phillips et al., 2021; Sampson & Levy, 2020).

We further found that the correlations among social dimensions were influenced by the severity of composite segregation. At the MSA level, the Global SSI-SES aligns most closely with the Global SSI-Comp. across all MSAs. Furthermore, SES↔HCD exhibited moderate coupling in MSAs with mid- to high composite segregation severity. In lightly segregated MSAs, HT↔SES and HT↔Comp. demonstrated moderate to strong coupling; however, this pattern did not persist in highly segregated areas. These findings can be explained in part by the strong correlation between SES and the Comp. index. They also suggest, however, that mobility-based segregation manifests through more complex mechanisms in highly segregated communities. Assessing segregation solely from a housing or residential perspective may therefore risk oversimplification. Such findings align with comparative frameworks emphasizing that segregation dimensions operate unevenly across contexts and scales (Reardon & Firebaugh, 2002; van Ham et al., 2021).

Last, OHSA and Jaccard analyses indicate that compounded segregation is rare but concentrated. The overlap of hot spots across social dimensions is generally low, with the highest observed pairwise similarity (SES-MSL) reaching 0.6 only in Phoenix. Nevertheless, a small share of tracts exhibits four-axis compounding; these "extreme" cases occur in only six MSAs (New York, Philadelphia, Los Angeles, Detroit, Portland, Dallas), led by New York (3.33% of tracts) and Philadelphia (1.92%). The rarity yet concentration of compounded segregation suggests that policies should (i) target these few extreme clusters with coordinated, multi-domain packages and (ii) elsewhere adopt axis-specific remedies rather than expecting cross-axis coincidence.

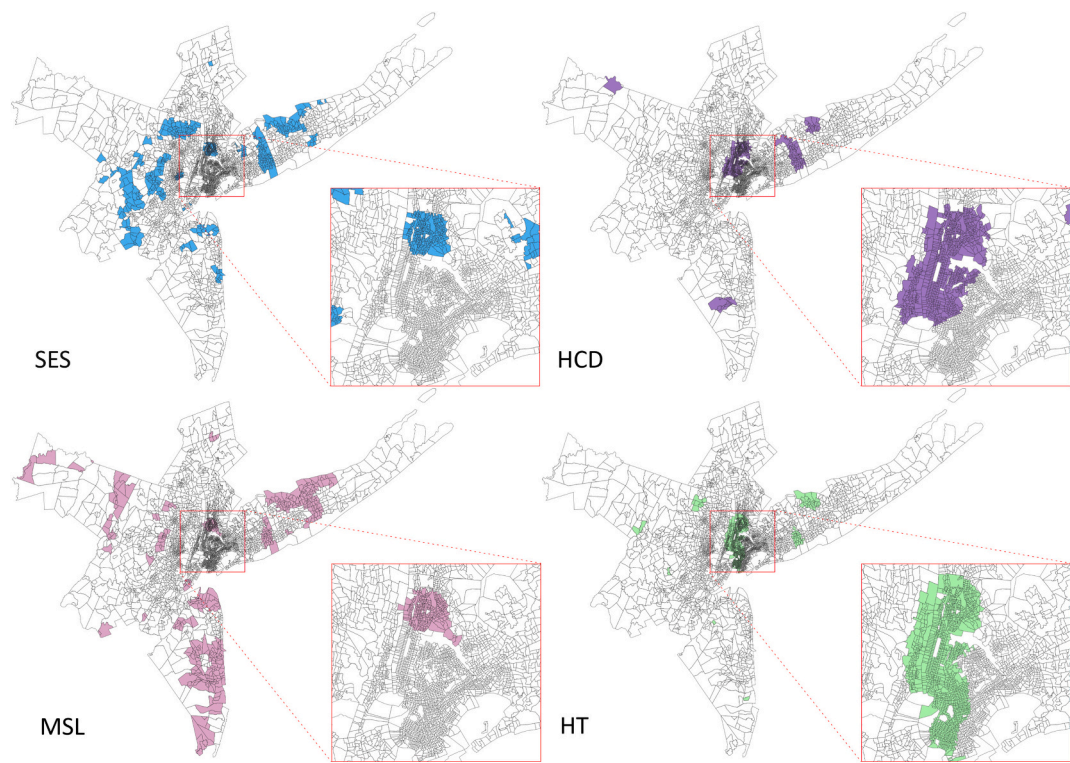


Fig. 8. Hot spot census tracts for the New York MSA.

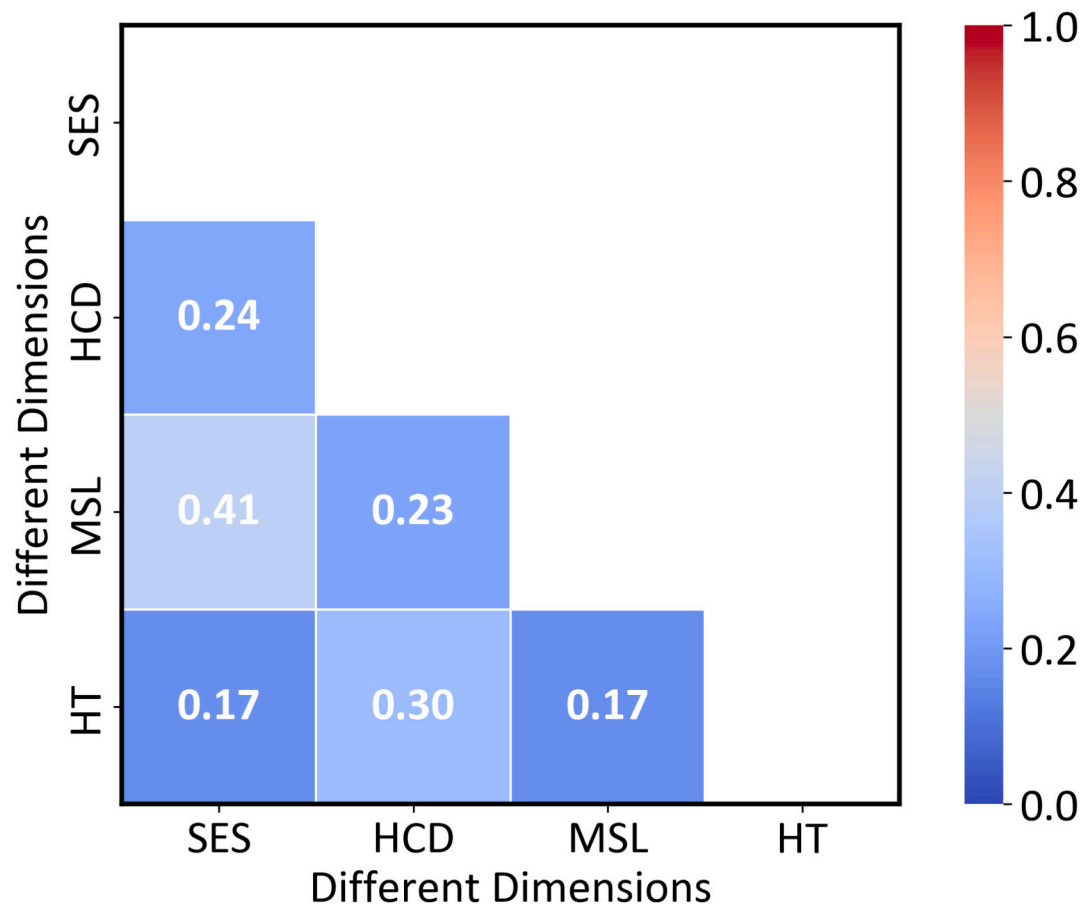


Fig. 9. Jaccard similarity results for hot spots identified across different dimensions in the New York MSA.

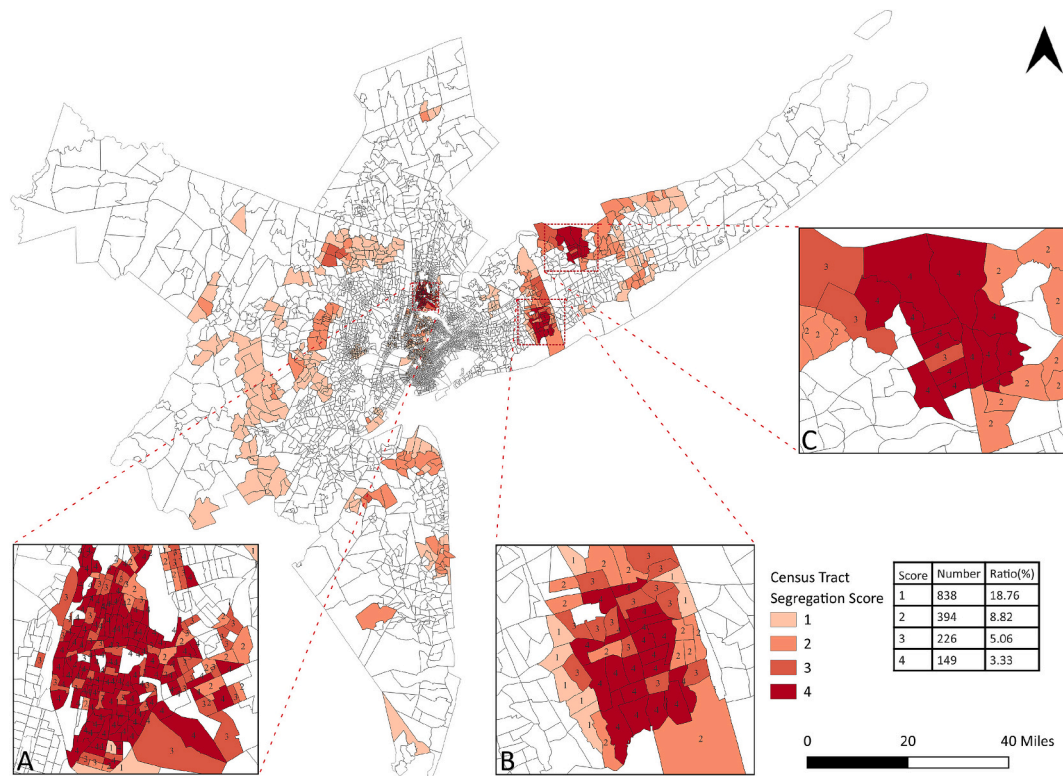


Fig. 10. Census tracts with compounded segregation issues in the New York MSA.

6.3. Limitations and future work

This study is subject to several limitations, which correspondingly illuminate productive pathways for future research.

From a conceptualization perspective, an important limitation in many existing studies is their normative interpretation of segregation. Although the term segregation is often associated with negative social outcomes, not all forms of spatial concentration or separation are inherently harmful. In this study, segregation is examined primarily as a manifestation of structural inequality, reflected in systematic differences in mobility-based exposure and access to opportunities across social groups. However, spatial homogeneity may also arise from voluntary processes rooted in homophily, shared culture, language, religion, or identity. Social network research has long demonstrated that homophilous interaction is a pervasive mechanism shaping social ties and can yield benefits such as reduced communication costs, stronger social support, and more stable interpersonal relations (McPherson et al., 2001). For some minority populations, co-ethnic residential clustering may offer protective effects by enhancing social support networks, mitigating exposure to discrimination, and strengthening a sense of belonging (Seo & Jivraj, 2025). Moreover, certain forms of residential clustering may be driven by the spatial concentration of commercial or cultural resources rather than exclusionary forces (Lee et al., 2025). Consequently, the results of this study should not be interpreted as implying that all spatial enclaves are problematic or should be dismantled through policy intervention. Instead, the policy relevance of our findings lies in identifying contexts where segregation reflects constrained mobility, unequal exposure, or limited access to opportunities. Future work could further differentiate between involuntary, structurally imposed segregation and voluntary forms of social clustering when

linking segregation patterns to normative or policy conclusions.

From a data perspective, our reliance on data aggregated to the census-tract and MSA scales means that all inferences are inherently collective or ecological in nature; consequently, they are not interpretable at the individual level and are susceptible to the ecological fallacy.

The use of composite CDC SVI themes, while convenient for aggregating diverse social indicators into distinct themes, introduces inherent collinearity among the indicators and hinders the nuanced exploration and separation of indicators that may have stronger spatial implications than social ones. For example, the percentage of households with no vehicle available is treated here as part of the HT theme, yet it may also shape accessibility and thus carry spatial implications. Additionally, the construction of the OD weights and the SSI is contingent on SafeGraph's device coverage, which may exhibit spatial and demographic biases, potentially leading to the undercounting of flows that cross MSA boundaries. Further efforts could be made to examine segregation from more granular social dimensions to further concretize the socio-spatial framework. Researchers may also incorporate data from multiple providers or leverage census-based commuting data to assess and address potential biases associated with SafeGraph's device coverage.

From a methodological perspective, this study adapts the segregation quantification method proposed by Xu et al. (2019), in which segregation is calculated using a relative similarity measure based on social ranks of OD pairs, rather than an indicator of absolute vulnerability levels. While this method facilitates comparison across individuals or cities by generating a consistent baseline value, the use of relative ranks may obscure differences between census tracts that have similar ranks but substantial differences in absolute values. Consequently, it may cause the method to overlook meaningful absolute disparities in vulnerability. Future efforts could consider integrating absolute

measures or standardized scores (e.g., z-scores) to provide a more complete picture of the segregation severity. In addition, our operationalization of compounded segregation is based on identifying extremely segregated census tracts, defined by the exact spatial overlap of high-severity segregation across multiple social dimensions at the census tract level. While this approach aligns with our objective of pinpointing highly segregated tracts, it does not explicitly account for spatial proximity or adjacency. Consequently, it fails to capture patterns of compounded segregation that may emerge from clusters of neighboring but non-overlapping tracts. Future research could extend this framework by incorporating proximity-sensitive approaches, such as distance-weighted overlap measures and spatially constrained similarity indices, which would enable researchers to differentiate between segregation concentrated within individual tracts and segregation that manifests as spatially clustered patterns, thereby providing a more comprehensive understanding of the spatial structure of compounded segregation.

7. Conclusions

This study, grounded in a mobility-based socio-spatial framework and powered by big mobility data across 30 MSAs in the U.S., compels a fundamental shift in how we measure and understand urban segregation. This study underscores that segregation is not a single condition but a multi-dimensional phenomenon. First, exposure-based measures such as Global/Local SSI, and clustering metrics of Moran's I and OHSA together capture different facets of the socio-spatial order, avoiding conflation of “more segregated” with “more clustered”. An MSA area can be highly segregated by exposure on an axis without showing strong clustering or vice versa. Second, the persistent volatility in city rankings across different social dimensions argues for a fundamental shift in analytical practice. Composite indices, while sometimes useful for summary, should be routinely accompanied by axis-disaggregated reporting to prevent the masking of critical, dimension-specific inequities. Third, regime-stratified correlation analysis should be standard practice. It identifies when dimension pairings emerge and prevents overgeneralizing an average correlation that masks regime shifts.

Appendix A. Appendix

Table A1
CDC SVI themes and variables.

CDC SVI Themes		CDC SVI Variables
Name	Description	Variable description
SES	Sum of series for the Socioeconomic theme	Percentage of persons below the poverty line Unemployment rate (age 16+) Per capita income
HCD	Sum of series for Household composition & disability theme	Percentage of persons with no high school diploma (age 25+) Percentage of persons aged 65 and older The percentage of persons who agreed 17 and younger Percentage of civilian noninstitutionalized population with a disability
MSL	Sum of series for Minority status & language barrier theme	Percentage of single-parent households with children under 18 Percentage minority (all persons except white, non-Hispanic)
HT	Sum of series for Housing type & transportation theme	Percentage of persons (age 5+) who speak English “less than well” Percentage of housing in structures with 10 or more units Percentage of mobile homes Percentage of occupied housing units with more people than rooms Percentage of households with no vehicle available
Comp.	Sum of all themes	Percentage of persons in group quarters

Recognizing this complexity enables both researchers and policymakers to move beyond composite indices, diagnose segregation more precisely, and design interventions aligned with the operative dimensions and mechanisms at play.

CRedit authorship contribution statement

Jinlin Shi: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Xiao Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Si Qiao:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Xiao Huang:** Writing – review & editing, Methodology, Data curation, Conceptualization. **Yang Xu:** Writing – review & editing, Methodology, Conceptualization. **Siqin Wang:** Writing – review & editing, Methodology. **Liangchen Zhou:** Writing – review & editing. **Guonian Lü:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Given his/her/their role as Editorial Board Member, Dr. Yang Xu had no involvement in the peer review of this article and had no access to information regarding its peer review. Full responsibility for the editorial process for this article was delegated to another journal editor. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table A2

The number and ratio of census tracts identified as hotspots from at least one dimension for each MSA.

MSA	Number of tracts	Ratio
Los Angeles-Long Beach-Anaheim (Los Angeles)	676	23.35
Las Vegas-Henderson-Paradise (Las Vegas)	113	23.20
San Francisco-Oakland-Berkeley (San Francisco)	216	22.22
Atlanta-Sandy Springs-Alpharetta (Atlanta)	191	20.19
New York-Newark-Jersey City (New York)	838	18.75
Washington-Arlington-Alexandria (Washington, DC)	251	18.59
Seattle-Tacoma-Bellevue (Seattle)	133	18.52
Chicago-Naperville-Elgin (Chicago)	406	18.43
Phoenix-Mesa-Chandler (Phoenix)	181	18.41
San Diego-Chula Vista-Carlsbad (San Diego)	112	18.00
Charlotte-Concord-Gastonia (Charlotte)	95	17.56
Detroit-Warren-Dearborn (Detroit)	217	16.90
Miami-Fort Lauderdale-Pompano Beach (Miami)	198	16.55
Portland-Vancouver-Hillsboro (Portland)	78	15.91
Philadelphia-Camden-Wilmington (Philadelphia)	210	14.37
Dallas-Fort Worth-Arlington (Dallas)	182	13.90
St. Louis (St. Louis)	69	11.23
Pittsburgh (Pittsburgh)	76	10.87
Houston-The Woodlands-Sugar Land (Houston)	97	9.10
Riverside-San Bernardino-Ontario (Riverside)	68	8.33
Cincinnati (Cincinnati)	41	8.20
Boston-Cambridge-Newton (Boston)	71	7.16
Sacramento-Roseville-Folsom (Sacramento)	32	6.60
Baltimore-Columbia-Towson (Baltimore)	40	5.95
Austin-Round Rock-Georgetown (Austin)	19	5.48
Tampa-St. Petersburg-Clearwater (Tampa)	39	5.31
San Antonio-New Braunfels (San Antonio)	20	4.41
Orlando-Kissimmee-Sanford (Orlando)	16	4.10
Minneapolis-St. Paul-Bloomington (Minneapolis)	17	2.17
Denver-Aurora-Lakewood (Denver)	11	1.79

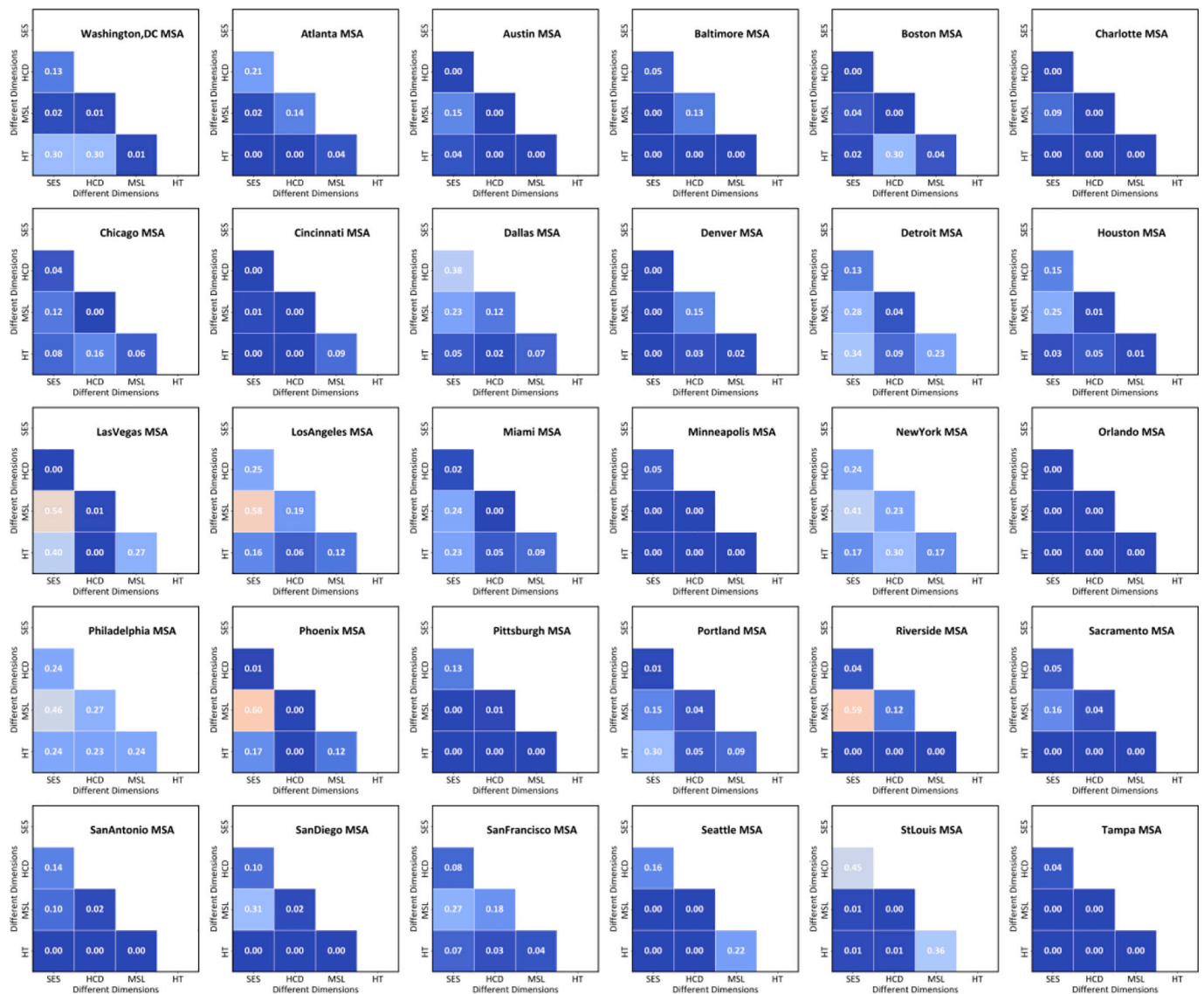


Fig. A1. Jaccard similarity results for hot spots identified across different dimensions in all 30 MSAs.

Data availability

The authors do not have permission to share data.

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